Course: ITA

PREDICTING VOLATILITY REGIMES IN THE NIFTY 50 USING MACHINE LEARNING

Volatility Regime Classification in the Indian Stock Market

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Section A

PROBLEM STATEMENT

Objective: Financial markets experience periods of high and low volatility. Identifying these regimes helps investors manage risk and adjust strategies.

Research Question

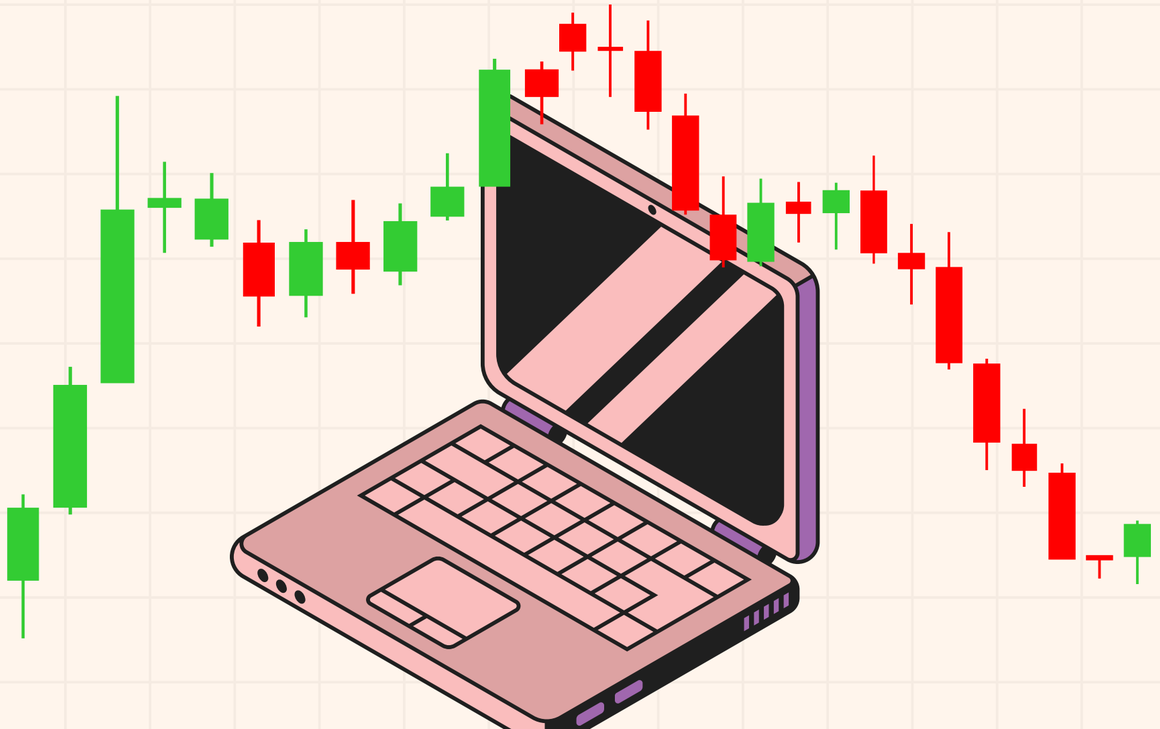
Can machine learning predict whether the market will enter a high-volatility or low-volatility regime?

Classification Task: Two regimes-

- High Volatility
- Low Volatility

Why this matters

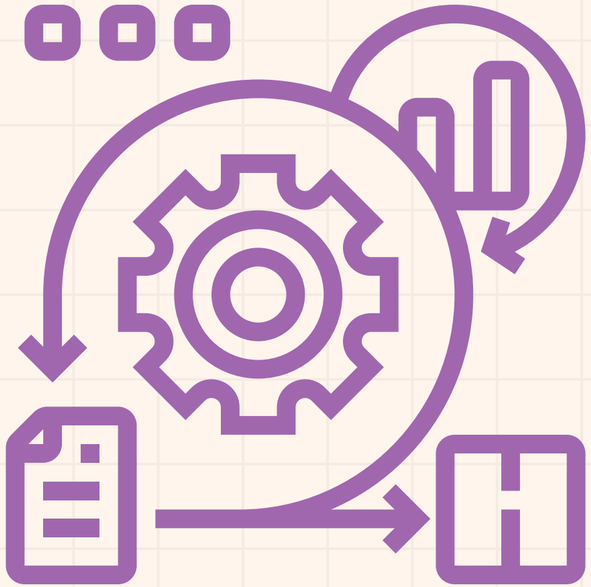
- Portfolio risk management
- Better asset allocation
- Improved trading strategies



DATASET & VARIABLES

Dataset (from yfinance library)

- Historical data from the Nifty 50
- Daily observations over multiple years



Key variables used

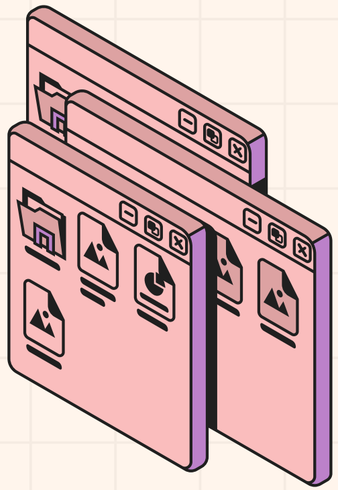
Market indicators such as:

- Momentum indicators
- Moving averages
- Trading volume
- Price returns
- Technical indicators

Volatility Measurement

Volatility calculated using rolling standard deviation of returns.

METHODOLOGY



Step 1: Data Preparation

- Calculate market returns
- Measure rolling volatility

Step 2: Regime Definition

Volatility regimes defined as:

- High Volatility: volatility above historical median
- Low Volatility: volatility below historical median

Step 3: Feature Construction

Indicators used include:

- Momentum indicators
- Trend indicators
- Volume indicators
- Risk indicators

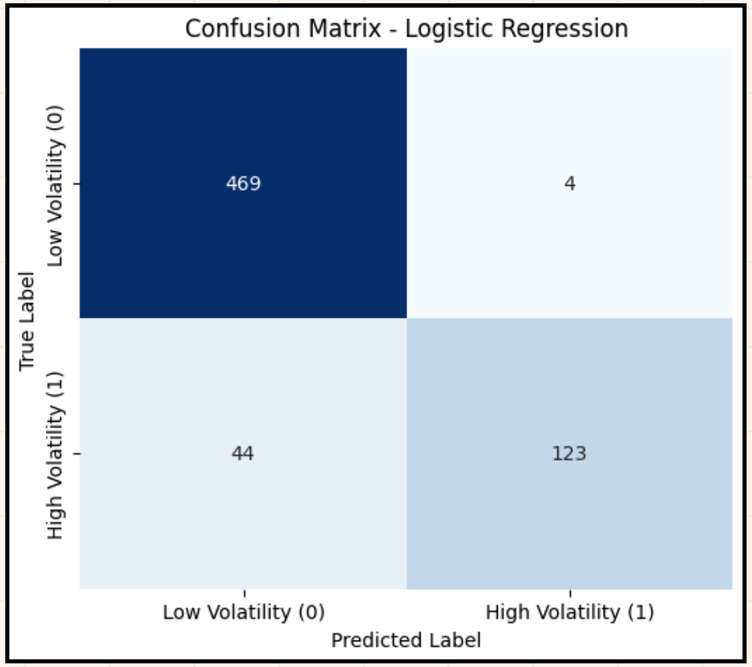
Step 4: Machine Learning Models

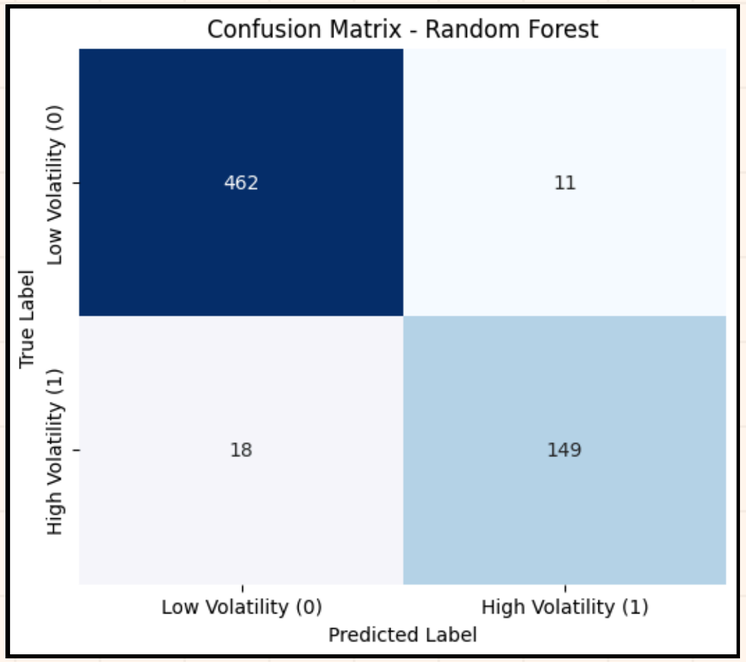
Several classification models were trained to predict volatility regimes.

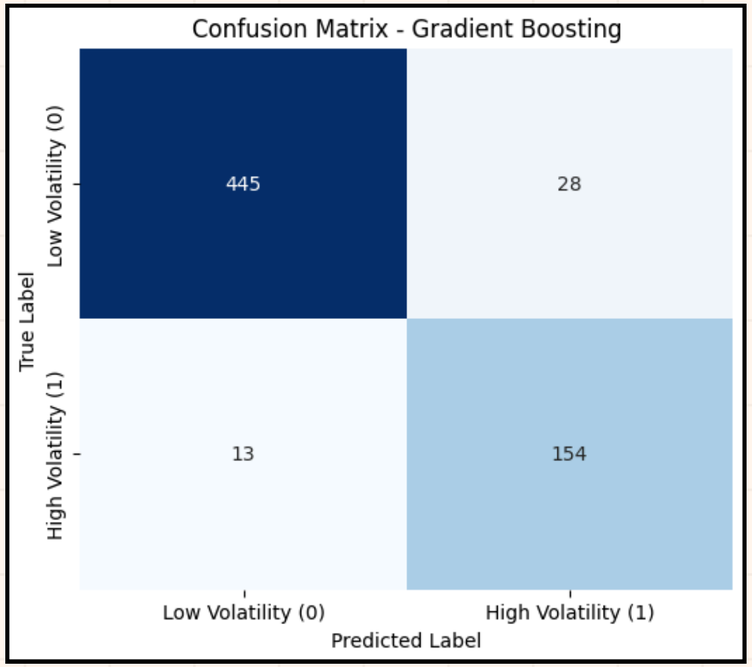
RESULTS

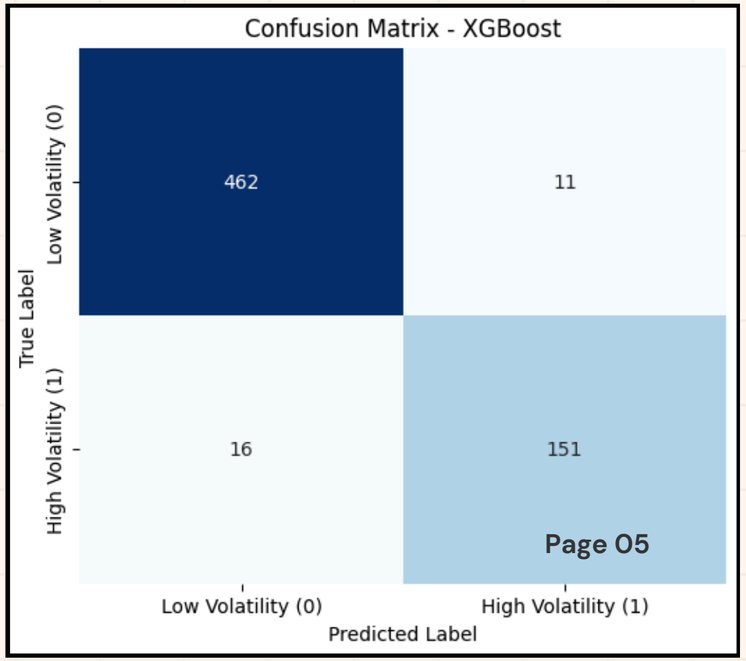
Trained and evaluated four machine learning models to classify volatility regimes (Low: 0, High: 1) on the Nifty 50 index.

- **Logistic Regression:**
  - Accuracy: 0.9250, Precision: 0.9685
  - Recall: 0.7365, F1-Score: 0.8367
  - ROC-AUC: 0.9829
- **Random Forest:**
  - Accuracy: 0.9547, Precision: 0.9313
  - Recall: 0.8922, F1-Score: 0.9113
  - ROC-AUC: 0.9787
- **Gradient Boosting:**
  - Accuracy: 0.9359, Precision: 0.8462
  - Recall: 0.9222, F1-Score: 0.8825
  - ROC-AUC: 0.9799
- **XGBoost:**
  - Accuracy: 0.9578, Precision: 0.9321
  - Recall: 0.9042, F1-Score: 0.9179
  - ROC-AUC: 0.9854

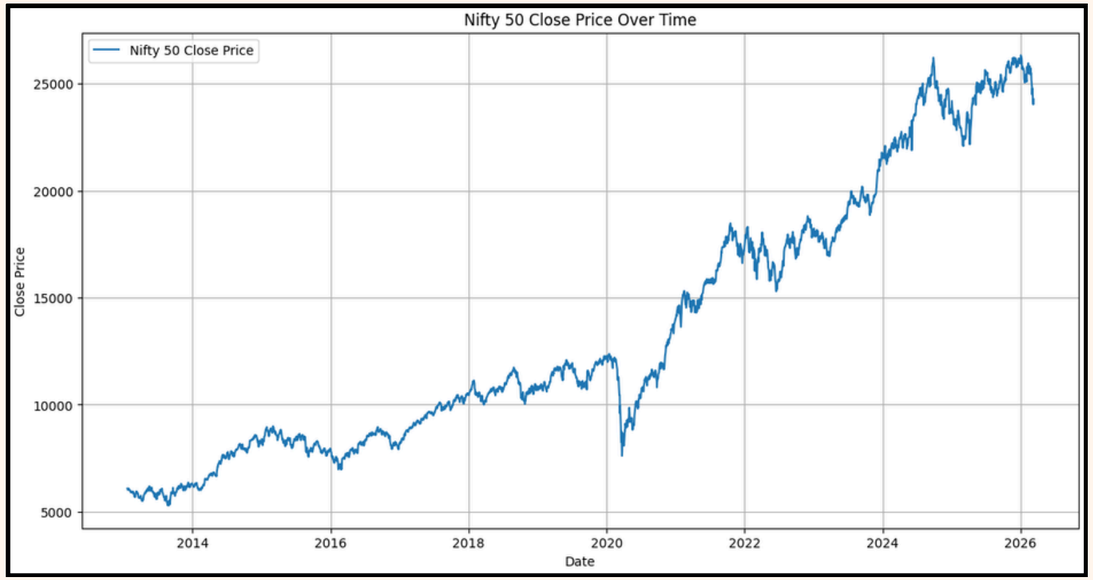


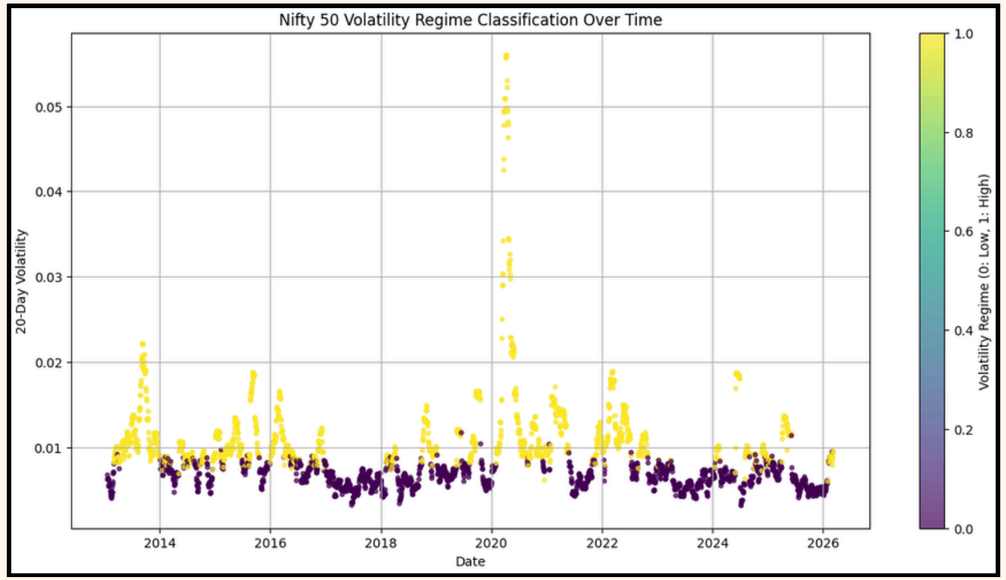


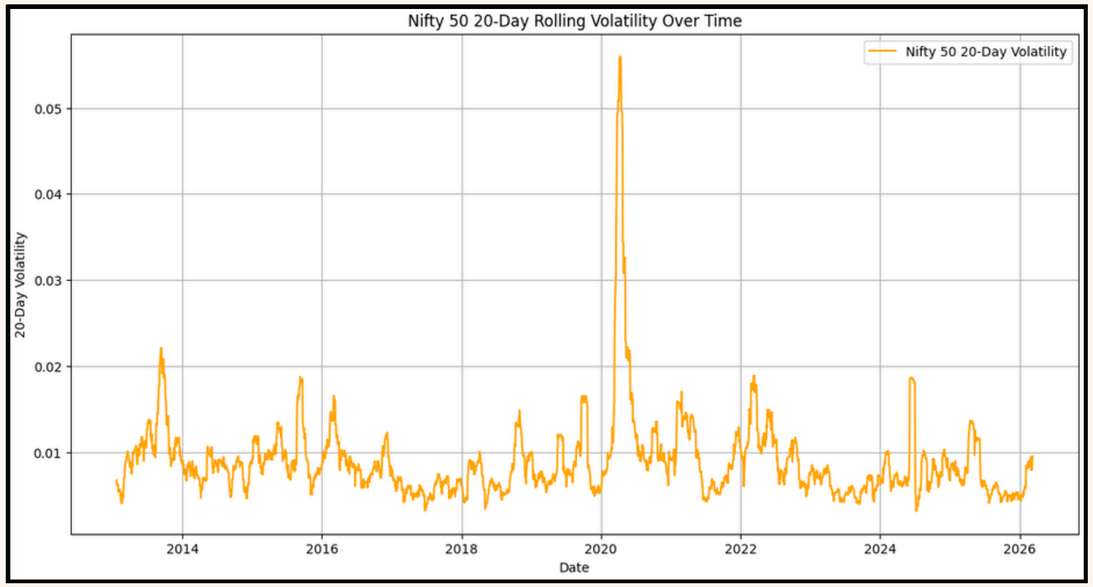




VISUALISATIONS

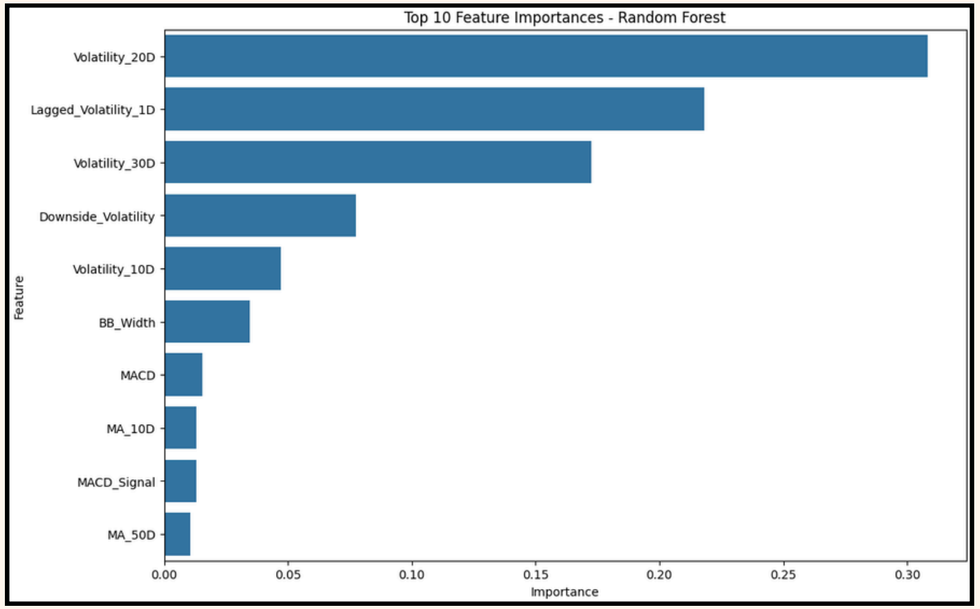


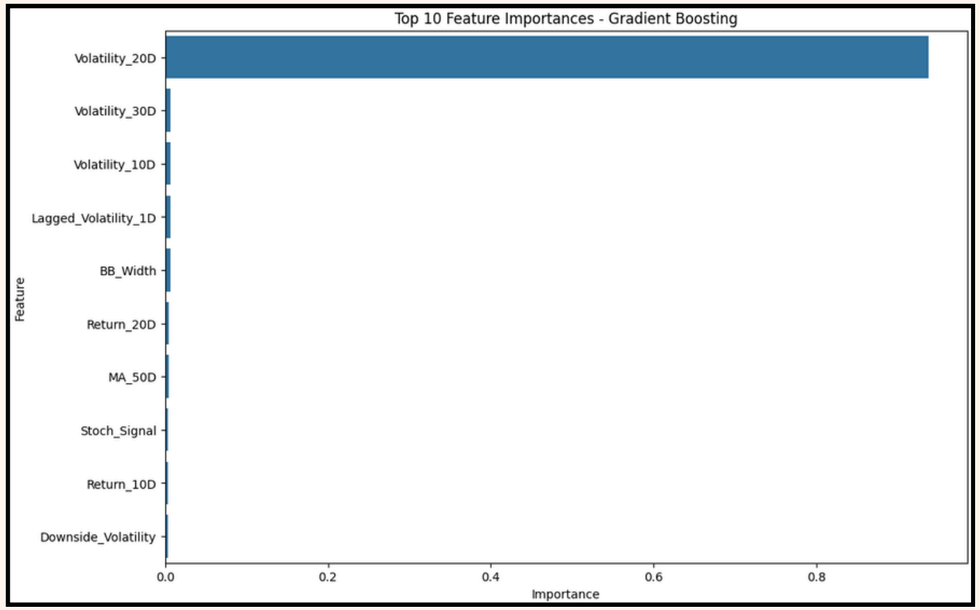


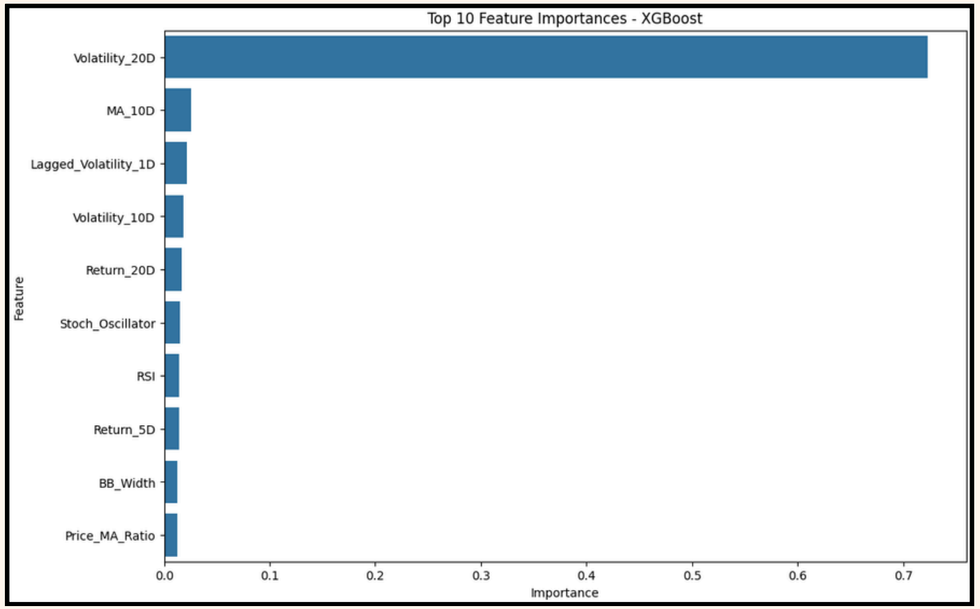


- Nifty 50 Close Price Over Time
- Nifty 50 20-Day Rolling Volatility Over Time
- Nifty 50 Volatility Regime Classification Over Time (Scatter Plot)
- Top 10 Feature Importances for Random Forest, Gradient Boosting, and XGBoost Models (PTO)

FEATURE IMPORTANCE







Extensive Feature Engineering: Created 22 predictive features, including:

- Momentum: 5, 10, 20-day returns.
- Trend: 10, 20, 50-day Moving Averages, Price/50-day MA ratio.
- Technical Indicators: RSI, MACD, Bollinger Band Width, Stochastic Oscillator.
- Volume Indicators: Volume Change, 10-day Volume MA, Relative Volume.
- Risk Indicators: Downside Volatility.
- Lagged Features: Lagged Daily Return, Lagged 20-day Volatility.

Best Performing Model

XGBoost emerged as the best-performing model for predicting Nifty 50 volatility regimes.

- It achieved the highest ROC-AUC (0.9854), indicating excellent discriminatory power.

It also had the highest F1-Score (0.9179), showing a strong balance between precision and recall.

Implications & Next Steps

- **Risk Management:** Predicting volatility regimes can significantly aid in dynamic risk management strategies, allowing for adaptive position sizing or hedging.
- **Algorithmic Trading:** This model could be integrated into trading algorithms to adjust strategies based on anticipated market volatility (e.g., trend-following in low volatility, mean-reversion in high volatility).

THANK YOU